

Here's why you need to add a scripting language to game engines:

- ▶ Scripting gives you the ability to code game's major functionality without having to compile the Game Engine.
- ▶ Those functionalities include Artificial Intelligence (AI), User Interface (UI), game events, save and load game functionality.
- ▶ Advantage to modders to change the look and feel of the game.
- ▶ Scripting allows people of a game company of various disciplines to work on their piece of code, without changing anything in the Game Engine.
- ▶ For example, say that a class for an object is coded in the game engine that can be placed in the game world. If a Level-Designer wants to place that object at a particular point in the game-world, hard-coding the coordinates of the world space and tweak the Game Engine will not only be tedious but also waste of time. Scripting in games would avoid this.



A split-screen game

Disadvantages of embedding scripting a game engine

- ▶ It is relatively slow as it runs on Virtual Machine.
- ▶ Scripts can't be used to create entirely new visual effects, because of lack of speed.
- ▶ People who are working on the Scripting Language have to learn the Language properly.

Embedding the script engine

- ▶ The Script Engine needs to be bound to the Game Engine so that a two-way communication can take place.
- ▶ The Script Engine code is embedded in the Game Engine code by linking explicitly or implicitly to a DLL or static .LIB library. This is not the only way to do, but it does represent the scenario in Angel Script and Lua embedding scenario.
- ▶ Hence, by this embedded script, the programmer can call the functions of